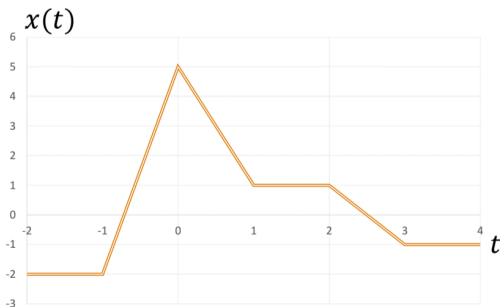


# Tarea 2

Tarea 2. Realizar las transformaciones indicadas para la siguiente señal y dibujar sus graficas correspondientes:

| t  | x(t) |
|----|------|
| -2 | -2   |
| -1 | -2   |
| 0  | 5    |
| 1  | 1    |
| 2  | 1    |
| 3  | -1   |
| 4  | -1   |



29 de 30

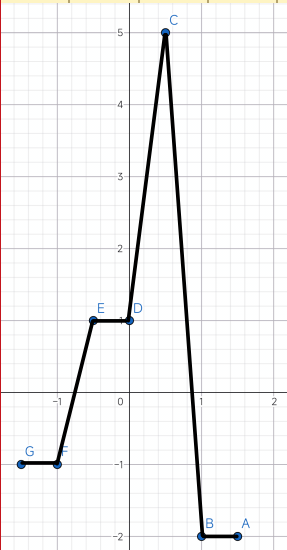
Obtener:

- a)  $x_1(t) = x(-2t + 1)$
- b)  $x_2(t) = 3x\left(\left(\frac{2}{5}\right)t - 2\right)$
- c)  $x_3(t) = \frac{1}{3}x(1.73t + \sqrt{2})$

a)  $x_1(t) = x(-2t + 1)$   
 $t_n = -2t_0 + 1$

$t_n - 1 = -2t_0$   
 $\frac{t_n - 1}{-2} = t_0$

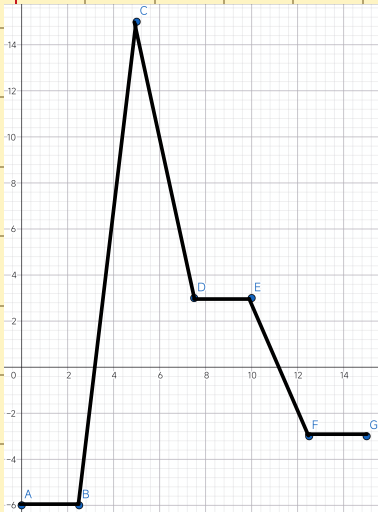
| $t_0$ | $x_1(t_0)$ |
|-------|------------|
| -1.5  | -2         |
| -1    | -2         |
| -0.5  | 5          |
| 0     | 1          |
| 0.5   | 1          |
| 1     | -1         |
| 1.5   | -1         |



b)  $x_2(t) = 3x\left(\left(\frac{2}{5}\right)t - 2\right)$

$t_n = \frac{2}{5}t_0 - 2$   
 $t_n + 2 = \frac{2}{5}t_0$   
 $\frac{5(t_n + 2)}{2} = \frac{5t_n + 10}{2} = \frac{5}{2}t_n + 5 = t_0$

| $t_0$ | $x_2(t_0)$ |
|-------|------------|
| 0     | -6         |
| 2.5   | -6         |
| 5     | 15         |
| 7.5   | 3          |
| 10    | 3          |
| 12.5  | -3         |
| 15    | -3         |



c)  $x_3(t) = \frac{1}{3}x(1.73t + \sqrt{2})$

$t_n = 1.73t_0 + \sqrt{2}$   
 $t_n - \sqrt{2} = 1.73t_0$   
 $t_0 = \frac{t_n - \sqrt{2}}{1.73}$

| $t_0$   | $x_3(t_0)$ |
|---------|------------|
| -1.9735 | -0.666     |
| -1.3954 | -0.666     |
| -0.9174 | 1.666      |
| -0.2394 | 0.333      |
| 0.3386  | 0.333      |
| 0.9166  | -0.333     |
| 1.4946  | -0.333     |

